Practical Problems

1. Find out the future value of Rs. 4,800 received after two years at 10% time preference rate.

2. Calculate the future value for 10,000 deposited in bank for a period of 5 years at 12% p.a. Given $(1.12)^5 = 1.762$

3. Calculate the future value of Rs. 38,000 invested for 5 years at a rate of interest of 15% compounded half yearly. According to compound table compound value factor for Rs. 1 in 5 years at rate 12%.

4. Calculate the future value of Rs. 18,000 is invested for a period of 5 years at 12% P.a. interest compound quarterly. Find out FV given $(1.03)^{20} = 1.806$

5. Calculate the future value of annuity of Rs. 66,000 deposited at the end of each year at 8% for a period of 5 years.

6. Mr. Satish Roy deposits Rs. 23,000 at the end of every year for five years and the deposit earns compound interest @10% p.a. Determine how much money he will have at the end of 5 years.

7. Calculate the future value at the end of 5 years of the following series of payments at 12% rate of interest.

   Rs. 2,000 at the end of first year.

   Rs. 4,000 at the end of second year.

   Rs. 6,000 at the end of third year.

   Rs. 8,000 at the end of fourth year.

   Rs. 10,000 at the end of fifth year.

8. Mr. Saha is to receive Rs. 12,000 after 5 years from now. His time preference for money is 12% p.a. Calculate the present value. According to discount factor table, the present value discount factor at 12% P.a. for 5 years for Rs. 1 is 0.567. Calculation of present value.

9. Calculate the future value of money at the end of five years of the following series of payments at 9% rate of interest.

   Rs. 22,000 at the end of first year.

   Rs. 23,000 at the end of second year.

   Rs. 24,000 at the end of third year.

   Rs. 26,000 at the end of fourth year.

10. Calculate the future value of money at the end of five years of the following series of payments at 5% rate of interest.

    Rs. 11,000 at the end of first year.

    Rs. 11,500 at the end of second year.

Rs. 12,000 at the end of third year.

Rs. 13,000 at the end of fourth year.

11. Find out the present value of annuity receipt of Rs. 7,000 received for 5 years at the rate of 12% discount rate.

12. Calculate the present value of the following series of payments made at the end of each year for a period of 5 years at 9% interest rate.

    Cash flow at the end of 1st year Rs. 1,000

    Cash flow at the end of 2nd year Rs. 2,000

    Cash flow at the end of 3rd year Rs. 3,000

    Cash flow at the end of 4th year Rs. 4,000

    Cash flow at the end of 5th year Rs. 5,000

13. Calculate the present value of the following series of payments made at the end of each year for a period of 4 years at 12% interest rate.

    Cash flow at the end of 1st year Rs. 11,000

    Cash flow at the end of 2nd year Rs. 12,000

    Cash flow at the end of 3rd year Rs. 13,000

    Cash flow at the end of 4th year Rs. 14,000

    Cash flow at the end of 5th year Rs. 15,000

14. Calculate the present value of the following series of payments made at the end of each year for a period of 6 years at 18% interest rate.

    Cash flow at the end of 1st year Rs. 5,000

    Cash flow at the end of 2nd year Rs. 6,000

    Cash flow at the end of 3rd year Rs. 7,000

    Cash flow at the end of 4th year Rs. 8,000

    Cash flow at the end of 5th year Rs. 9,000

15. Calculate the present value of the following series of payments made at the end of each year for a period of 4 years at 6% interest rate.

    Cash flow at the end of 1st year Rs. 2,000

    Cash flow at the end of 2nd year Rs. 4,000

    Cash flow at the end of 3rd year Rs. 6,000

    Cash flow at the end of 4th year Rs. 8,000

    Cash flow at the end of 5th year Rs. 10,000

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